

Seong-Kwan Hong

researcher

基本信息

姓名: Seong-Kwan Hong

所属院校: 汉阳大学

地区: South Korea

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

混合信号集成技术

模拟与数模混合电路

模拟与射频IC设计

超大规模集成电路设计

低功耗电路与能量管理

电力电子与电机驱动

微电子器件与制造

光电子器件与微纳光学

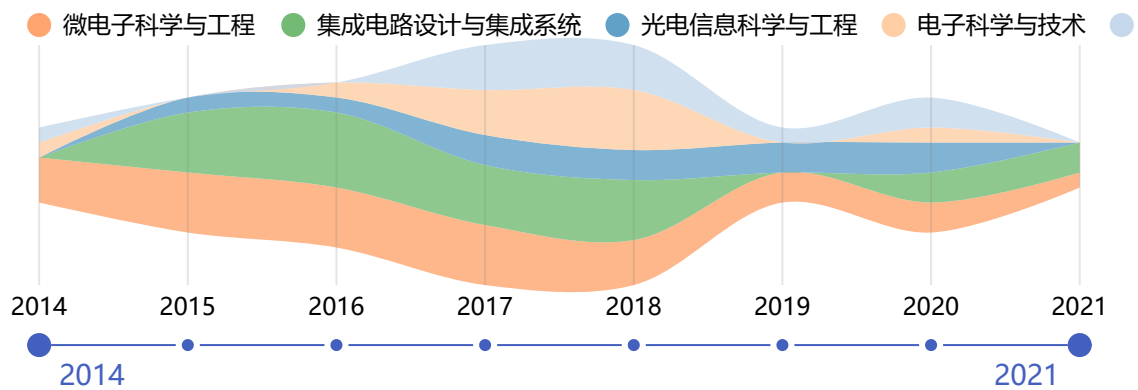
智能功率集成电路

光电显示与新型光源

个人简介

Seong-Kwan Hong is an academic researcher from Hanyang University. He has contributed to research in topics such as Integral nonlinearity and CMOS. With an h-index of 12, he has co-authored 65 publications. His works include research on display drivers, pixel circuits, touch systems, and analog-to-digital converters, published in well - known IEEE journals like IEEE Transactions on Circuits and Systems, IEEE Electron Device Letters, etc.

近年发表的论文



A High Frame Rate Analog Front-End IC With Piezoelectric Micromachined Ultrasound Transducers Using Analog Multi-Line Acquisition for Ultrasound Imaging Systems

IEEE ACCESS

Jeong, Ji-Yong; Song, Jong-Keun; Choi, Min-Seog; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 0

2021

A High-Speed and Energy-Efficient Multi-Bit Cyclic ADC Using Single-Slope Quantizer for CMOS Image Sensors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jeong, Ji-Yong; Shim, Junbo; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 6

2021

A Highly Reliable SIMO Converter Using Hybrid Starter and Overcharging Protector for Energy Harvesting Systems

IEEE ACCESS

Jung, Jae-Hyung; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 1

2020

An AMOLED Pixel Circuit With a Compensating Scheme for Variations in Subthreshold Slope and Threshold Voltage of Driving TFTs

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Keum, Nack-Hyeon; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 28

2020

A Temperature Compensation Method by Adjusting Gamma Voltages for High Luminance Uniformity of Active Matrix Organic Light-Emitting Diode Displays

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Na, Jun-Seok; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 5

2020

A High Peak Output Power and High Power Conversion Efficiency SIMIMO Converter Using Optimal on-Time Control and Hybrid Zero Current Switching for Energy Harvesting Systems in IoT Applications

IEEE TRANSACTIONS ON POWER ELECTRONICS

Jung, Jae-Hyung; Jung, Young-Ho; Hong, Seong-Kwan; Kwon, Oh-Kyong

A Highly Linear 10-Bit DAC of Data Driver IC Using Source Degeneration Load for Active Matrix Flat-Panel Displays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Na, Jun-Seok; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 10

2020

A Fast Transient Response Hybrid LDO With Highly Accurate DC Voltage Using Countable Bidirectional Binary Search and Soft Swap Switching

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jung, Jae-Hyung; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 12

2020

A Fast Transient and Low Power Capacitor-less Low-dropout Regulator using Advanced Dynamic Biasing Method

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Jung, Jae-Hoon; Jung, Jae-Hyung; Jung, Young-Ho; Jeong, Hoe-Eung; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 0

2019

A 4410-ppi Resolution Pixel Circuit for High Luminance Uniformity of OLEDs Microdisplays

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Na, Jun-Seok; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 18

2019

A Compensation Method for Variations in Subthreshold Slope and Threshold Voltage of Thin-Film Transistors for AMOLED Displays

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Keum, Nack-Hyeon; Chai, Chong-Chul; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 7

2019

An Ultra-Low-Power 16-Bit Second-Order Incremental ADC With SAR-Based Integrator for IoT Sensor Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Shim, Junbo; Kim, Min-Kyu; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 10

2018

A Highly Power-Efficient Single-Inductor Bipolar-Output DC-DC Converter Using Hysteretic Skipping Control for OLED-on-Silicon Microdisplays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kwak, Bong-Choon; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 15

2018

A Low-Power 12-Bit Extended Counting ADC Without Calibration for CMOS Image Sensors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jeon, Byoung-Kwan; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 12

2018

A Low-Power Analog Delay Line Using a Current-Splitting Method for 3-D Ultrasound Imaging Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jeong, Ji-Yong; An, Jae-Sung; Jung, Sung-Jin; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 4

2018

An Active Matrix Micro-Pixelated LED Display Driver for High Luminance Uniformity Using Resistance Mismatch Compensation Method

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ahn, Hyun-A; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 52

2018

An AMOLED pixel circuit for high image quality of 1000ppi mobile displays in AR and VR applications

JOURNAL OF THE SOCIETY FOR INFORMATION DISPLAY

Keum, Nack-Hyeon; Hong, Seong-Kwan; Chai, Chong Chul; Kwon, Oh-Kyong

被引用次数: 8

2018

A Highly Accurate Current LED Lamp Driver With Removal of Low-Frequency Flicker Using Average Current Control Method

IEEE TRANSACTIONS ON POWER ELECTRONICS

Ahn, Hyun-A; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 19

2018

A Fast and Highly Accurate Battery Charger With Accurate Built-In Resistance Detection

IEEE TRANSACTIONS ON POWER ELECTRONICS

Jung, Young-Ho; Jung, Jae-Hyung; Jeong, Hoe-Eung; Jung, Jae-Hoon; An, Jae-Sung; Ahn, Hyun-A.; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 14

2018

Capacitive Touch Systems With Styli for Touch Sensors: A Review

IEEE SENSORS JOURNAL

Kwon, Oh-Kyong; An, Jae-Sung; Hong, Seong-Kwan

被引用次数: 41

2018

A Highly Linear and Accurate Fork-Shaped Electrode Pattern for Large-Sized Capacitive Touch Screen Panels

IEEE SENSORS JOURNAL

Lee, Seung-Hwan; An, Jae-Sung; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 8

2018

A 3.9-kHz Frame Rate and 61.0-dB SNR Analog Front-End IC With 6-bit Pressure and Tilt Angle Expressions of Active Stylus Using Multiple-Frequency Driving Method for

Capacitive Touch Screen Panels

IEEE JOURNAL OF SOLID-STATE CIRCUITS

An, Jae-Sung; Han, Sang-Hyun; Kim, Ju Eon; Yoon, Dong-Hyun; Kim, Young-Hwan; Hong, Han-Hee; Ye, Jae-Hun; Jung, Sung-Jin; Lee, Seung-Hwan; Jeong, Ji-Yong; Baek, Kwang-Hyun; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 23

2018

A fully integrated switched-capacitor DC-DC converter with hybrid output regulation

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Chung, Kyunghoon; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 1

2018

A fast transient LED driver with adaptive frequency control according to load variation for large-sized LCD backlights

JOURNAL OF THE SOCIETY FOR INFORMATION DISPLAY

Jung, Young-Ho; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 1

2017

A high-efficient and fast-transient buck-boost converter using adaptive direct path skipping and on-duty modulation

MICROELECTRONICS JOURNAL

Jung, Young-Ho; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 4

2017

A low load- and cross-regulation SIDO converter using an adaptive current sensor and LDO regulator with a selectable charge pump for mobile devices

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Jung, Young-Ho; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 3

2017

A small-area and low-power data driver IC using two-stage DAC with a capacitor array for active matrix flat-panel displays

JOURNAL OF THE SOCIETY FOR INFORMATION DISPLAY

Seol, Hyeon-Cheon; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 2

2017

A CMOS-based Temperature Sensor with Subthreshold Operation for Low-voltage and Low-power On-chip Thermal Monitoring

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Na, Jun-Seok; Shin, Woosul; Kwak, Bong-Choon; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 3

2017

A Tileable CMOS X-Ray Line Detector Using Time-Delay-Integration With Pseudomultisampling for Large-Sized Dental X-Ray Imaging Systems

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jo, Yun-Rae; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 8

2017

An AMOLED Panel Test System Using Universal Data Driver ICs for Various Pixel Structures

IEEE TRANSACTIONS ON ELECTRON DEVICES

Seol, Hyeon-Cheon; Ra, Jong-Hyun; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 7

2017

Low-Power Low-Noise Amplifier Using Attenuation-Adaptive Noise Control for Ultrasound Imaging Systems

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Jung, Sung-Jin; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 14

2017

A Highly Noise-Immune Capacitive Touch Sensing System Using an Adaptive Chopper Stabilization Method

IEEE SENSORS JOURNAL

An, Jae-Sung; Jung, Sung-Jin; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 21

2017

A Driving Method of Pixel Circuit Using a-IGZO TFT for Suppression of Threshold Voltage Shift in AMOLED Displays

IEEE ELECTRON DEVICE LETTERS

Shin, Woo-Sul; Ahn, Hyun-A; Na, Jun-Seok; Hong, Seong-Kwan; Kwon, Oh-Kyong; Lee, Ji-Hun; Um, Jea-Gwang; Jang, Jin; Kim, Sung-Hwan; Lee, Jeong-Soo

被引用次数: 41

2017

A 70 MHz Temperature-Compensated On-Chip CMOS Relaxation Oscillator for Mobile Display Driver ICs

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Chung, Kyunghoon; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 2

2016

A Readout IC Using Two-Step Fastest Signal Identification for Compact Data Acquisition of PET Systems

SENSORS

Jung, Sung-Jin; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 3

2016

A Highly Linear and Accurate Touch Data Extraction Algorithm Based on Polar Coordinates for Large-Sized Capacitive Touch Screen Panels

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS

An, Jae-Sung; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 18

2016

A Fast Multiple Sampling Method for Low-Noise CMOS Image Sensors With Column-Parallel 12-bit SAR ADCs

SENSORS

Kim, Min-Kyu; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 8

2016

Simple pixel circuits for high resolution and high image quality organic light emitting diode-on-silicon microdisplays with wide data voltage range

JOURNAL OF THE SOCIETY FOR INFORMATION DISPLAY

Hong, Sang-Woon; Kwak, Bong-Choon; Na, Jun-Seok; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 11

2016

A Low-Noise and Area-Efficient PWM- $\Delta\Sigma$ ADC Using a Single-Slope Quantizer for CMOS Image Sensors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jo, Yun-Rae; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 18

2016

An Area-Efficient and Low-Power 12-b SAR/Single-Slope ADC Without Calibration Method for CMOS Image Sensors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Min-Kyu; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 47

2016

A Low-Power Two-Tap Voltage-Mode Transmitter With Precisely Matched Output Impedance Using an Embedded Calibration Circuit

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kwak, Kang-Sub; Ra, Jong-Hyun; Moon, Ho-Sung; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 8

2016

A Fast Switching Current Regulator Using Slewing Time Reduction Method for High Dimming Ratio of LED Backlight Drivers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ahn, Hyun-A; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 12

2016

High-speed two-step single-slope ADC using multi-sampling with partial conversion

ELECTRONICS LETTERS

Kim, Jong-Boo; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 4

2015

A High-Speed Wafer-Scale CMOS X-Ray Detector With Column-Parallel ADCs Using Oversampling Binning Method

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Jong-Boo; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 5

2015

Lifetime Extension Method for Active Matrix Organic Light-Emitting Diode Displays Using a Modified Stretched Exponential Decay Model

IEEE ELECTRON DEVICE LETTERS

Oh, Kyonghwan; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 11

2015

A Low-Power CMOS Image Sensor With Area-Efficient 14-bit Two-Step SA ADCs Using Pseudomultiple Sampling Method

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Jong-Boo; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 33

2015

A Small-Area and Energy-Efficient 12-bit SA-ADC With Residue Sampling and Digital Calibration for CMOS Image Sensors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Min-Kyu; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 9

2015

A Multi-Bit Incremental ADC Based on Successive Approximation for Low Noise and High Resolution Column-Parallel Readout Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jo, Yun-Rae; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 17

2015

CMOS Flat-Panel X-ray Detector With Dual-Gain Active Pixel Sensors and Column-Parallel Readout Circuits

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Jo, Yun-Rae; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 12

2014

Process Optimization of Integrated SiCr Thin-Film Resistor for High-Performance Analog Circuits

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kwon, Young-Cheon; Seol, Hyeon-Cheon; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 12

2014

Area-efficient high-voltage switch using floating control circuit for 3D ultrasound imaging systems

ELECTRONICS LETTERS

Jung, Sung-Jin; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 12

2014

5 Gbit/s 2-tap low-swing voltage-mode transmitter with least segmented voltage-mode equalisation

ELECTRONICS LETTERS

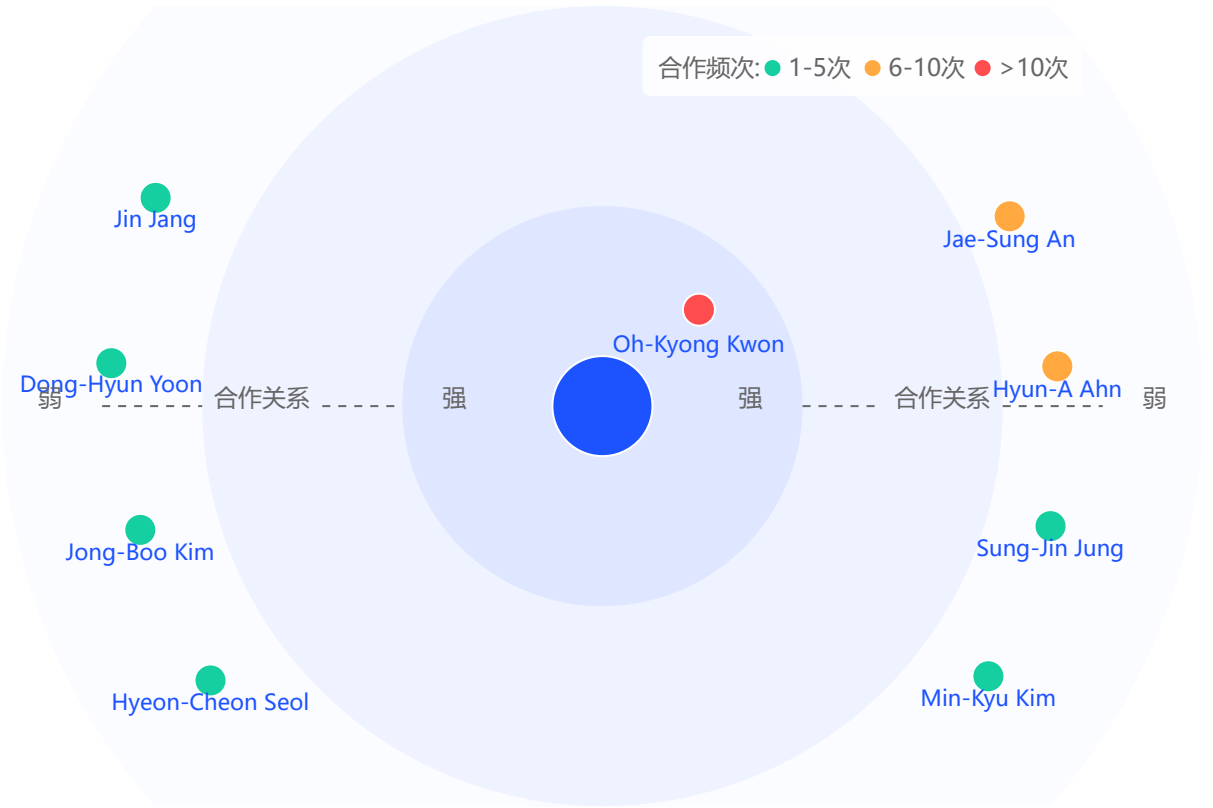
Kwak, Kang-Sub; Hong, Seong-Kwan; Kwon, Oh-Kyong

被引用次数: 6

2014

A Highly Power-Efficient Single-Inductor Multiple-Outputs (SIMO) DC-DC Converter with Gate Charge Sharing Method

学术合作



Oh-Kyong Kwon	合作次数: 50
汉阳大学	
Jae-Sung An	合作次数: 7
代尔夫特理工大学	
Hyun-A Ahn	合作次数: 6
汉阳大学	
Sung-Jin Jung	合作次数: 5
汉阳大学	
Min-Kyu Kim	合作次数: 4
汉阳大学	

还有 4 个合作者